

Chapter 5 Dimensional analysis and similarity principle

Section 1 Dimension

Section 2 Method of dimensional analysis

Section 3 Basics of flow similarity theory

Section 4 Model experiment

Chapter Summary

Chapter 5 Dimensional analysis and similarity principle

Key study points:

- Significance of dimensional analysis; principle of dimensional homogeneity
- Method of dimensional analysis: Rayleigh method and π theorem
- Similarity principle: conditions of similarity; similarity criterion number; dynamical similarity criterion



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Chapter 5 Dimensional analysis and similarity principle

Section 1 Dimension



1. Basic concepts
2. Principle of dimensional homogeneity

Section 1 Dimension

(1) Basic concepts of dimension

Unit: A standard quantity chosen for measuring the amount of a certain physical variable. E.g. the unit of length is m or cm.

Dimension: Indicating the property and category of a certain physical variable, without considering the unit. E.g. the dimension of length is [L].

Dimension {

- Fundamental dimension (基本量纲):** Dimensions that are independent and cannot be deduced from other dimensions. Length, mass and time are always taken. [LMT]
- Derived dimension (导出量纲):** that derived from **fundamental dimensions**
E.g.: velocity, $\dim v = LT^{-1}$; acceleration, $\dim a = LT^{-2}$;
 force, $\dim F = MLT^{-2}$; dynamic viscosity, $\dim \mu = ML^{-1} T^{-1}$



(2) Dimensional formula

$$\dim q = L^{\alpha} T^{\beta} M^{\gamma}$$

Classification {

- Geometric dimension: $\alpha \neq 0, \beta = 0, \gamma = 0$
- Kinematic dimension: $\alpha \neq 0, \beta \neq 0, \gamma = 0$
- Dynamic dimension: $\alpha \neq 0, \beta \neq 0, \gamma \neq 0$

Quantities of dimension one (量纲一的量; dimensionless number, pure number, e.g., **similarity criterion number**): $\alpha = 0, \beta = 0, \gamma = 0$, so $\dim q = 1$, e.g., π, θ and **Re**, etc.

$$Re = \frac{vd}{\nu}, \quad \dim Re = \frac{(LT^{-1})}{L^2T^{-1}} = 1$$

Features: 1) The quantity of a dimensionless parameter/number is independent on the unit adopted;

2) No scale effect, i.e., with the characteristics of similarity criterion number (相似准数);

3) Transcendental function (超越函数) (e.g., logarithmic, exponential, trigonometric) operations can be performed.



Chapter 5 Dimensional analysis and similarity principle

Section 1 Dimension

1. Basic concepts

 2. Principle of dimensional homogeneity
(量纲和谐原理)

(1) Principle of dimensional homogeneity (量纲和谐原理)

- For all physical equations that correctly reflect the objective laws, the dimensions of all terms in the equations must be consistent.
- That is, an equation is valid only when the dimensions on both sides are the same.
- This is called the principle of dimensional homogeneity.

For example
$$z_1 + \frac{p_1}{\rho g} + \frac{\alpha_1 v_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{\alpha_2 v_2^2}{2g} + h_w$$

Importance of the principle:

- a. For checking the correctness and completeness of empirical formula
- b. For determining the index of physical quantities in formula
- c. For establishing the basic form of physical equations

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Section 1 Dimension



Section 2 Method of dimensional analysis

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Section 2 Method of dimensional analysis

-  1. Rayleigh Method
- 2. Buckingham π Theorem

Section 2 Dimensional analysis method

(1) Rayleigh Method

A direct application of the principle of dimensional homogeneity

(2) Calculation steps

- 1) Determine n **physical variables** related to the physical phenomena studied;
- 2) Write the **exponential product** of all the physical variables, such as:

$$F_D = k D^a U^b \rho^c \mu^d$$

- 3) Express the dimensions of physical variables by fundamental dimensions in the exponential product;
- 4) Determine the **indexes a, b, c, d** of physical variables according to the principle of dimensional homogeneity, and substitute the indexes to get the relationship among dimensions

Scopes of application: In general, the number of relevant unknowns should be **less than or equal to 4**.



Example 1: Find the formulation for Pitot flow velocity meter

Solution: (1) Determine the variables related to the physical phenomena studied.

Flow velocity u ; Pressure difference Δp between the dynamic pressure orifice A and static pressure orifice B of the Pitot tube; Fluid density ρ ; Gravity acceleration g .

That is: $u = f(\Delta p, \rho, g)$

(2) Write the exponential product of physical quantities

$$u = k \Delta p^a \rho^b g^c$$

Applying the [M-L-T] system and substituting the corresponding dimensions gives:

$$LT^{-1} = (ML^{-1}T^{-2})^a (ML^{-3})^b (LT^{-2})^c$$

(3) Determining the indexes of physical quantities according to the dimensional homogeneity principle:

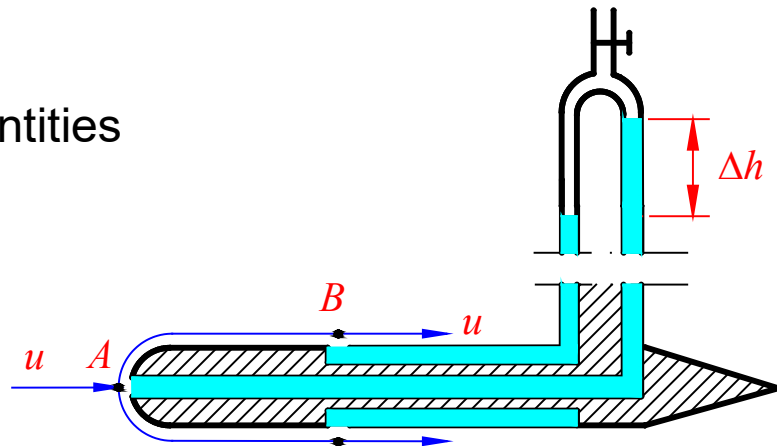
$$M: 0 = a + b$$

$$L: 1 = -a - 3b + c \quad \Rightarrow \quad a = 0.5, b = -0.5, c = 0$$

$$T: -1 = -2a - 2c$$

(4) Write the relationship among the related physical quantities:

$$u = k \sqrt{\frac{\Delta p}{\rho}} \quad \text{or} \quad u = c \sqrt{2g\Delta h}$$



Example 2: Determine the expression τ_0 for the wall shear stress in a circular tube

(1) Determine variables related to the physical phenomena:

Fluid density ρ ; dynamic viscosity coefficient μ ; Tube diameter D ; Wall roughness Δ ; Mean flow velocity v

(2) Write the exponential product of physical quantities

$$\tau_0 = K \cdot \rho^a \cdot \mu^b \cdot D^c \cdot \Delta^d \cdot v^e$$

Substituting dimensions gives

$$[M \cdot L^{-1} \cdot T^{-2}] =$$

$$[M \cdot L^{-3}]^a \cdot [M \cdot L^{-1} \cdot T^{-1}]^b \cdot [L]^c \cdot [L \cdot T^{-1}]^e \cdot [L]^d$$

(3) Determining the indexes of physical quantities according to the dimensional homogeneity principle:

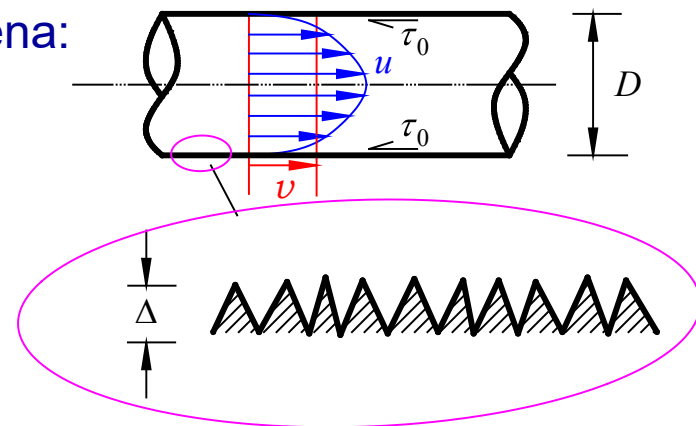
$$\left. \begin{array}{l} [M]: \quad 1 = a + b \\ [L]: \quad -1 = -3a - b + c + d + e \\ [T]: \quad -2 = -b - e \end{array} \right\}$$

Solving the above equations gives:

$$b=1-a, c=a-d-1, e=a+1$$

(4) Write the relationship among the related physical quantities

$$\tau_0 = K \cdot \rho^a \cdot \mu^{1-a} \cdot D^{a-d-1} \cdot \Delta^d \cdot v^{1+a}$$



Re-arranging gives

$$\tau_0 = K \left(\frac{\rho D v}{\mu} \right)^a \cdot \left(\frac{\Delta}{D} \right)^d \cdot \left(\frac{\mu}{\rho D v} \right) \cdot \rho v^2$$

$$= K (Re)^{a-1} \cdot \left(\frac{\Delta}{D} \right)^d \cdot \rho v^2$$

$$= f(Re, \frac{\Delta}{D}) \rho v^2$$

$$\text{Let } f(Re, \frac{\Delta}{D}) = \frac{\lambda}{8}$$

where λ is a coefficient determined by experiment.

$$\tau_0 = \frac{\lambda}{8} \rho v^2$$

Chapter 5 Dimensional analysis and similarity principle

Section 2 Method of dimensional analysis

1. Rayleigh Method

 2. Buckingham π Theorem

(1) Definition of π theorem

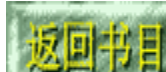
For a physical phenomenon, if there are n variables that are functions of each other, i.e. $F(x_1, x_2, \dots, x_n) = 0$. If containing m fundamental variables, these variables can be expressed as a function $\varphi(\pi_1, \pi_2, \dots, \pi_{n-m}) = 0$ in terms of $n-m$ dimensionless variables. That is, the n variables can be converted into $n-m$ dimensionless π numbers.

(2) Solution steps of π Theorem:

1) **Determine the general form of the function:** Based on physics, identify the physical variables and write down their relationships:

$$f(x_1, x_2, \dots, x_n) = 0$$

2) **Determine the fundamental variables:** Select m (generally $m = 3$) fundamental variables from the n physical quantities as the representatives of fundamental dimensions. For pipe flow, d , v and ρ are generally selected as the fundamental variables, while for open-channel flow, H , v , ρ are always selected.



(2) Solution steps of π Theorem (Cont'd)

3) Determine the number of π numbers $N(\pi) = n - m$. Write the π formulation consisting of the remaining physical quantities and the fundamental variables:

$$\pi_i = x_1^{a_i} x_2^{b_i} x_3^{c_i} \cdots x_i \quad (i = 1, 2, \cdots n - m)$$

4) Determine the π numbers of dimension one: Solve the equation system of indexes of variables in π numbers, to get the indexes a, b and c. With these, the π numbers of dimension one are determined.

5) Substitute all π terms into the π —theorem function

$$f(\pi_1, \pi_2, \cdots \pi_{n-m}) = 0$$

Or get a π parameter, such as:

$$\pi_4 = f(\pi_1, \pi_2, \cdots \pi_{n-m})$$

Or obtain a function for the dependent variable.



(3) Principles in selecting fundamental variables

- 1) Fundamental variables correspond to fundamental dimensions. If three fundamental dimensions (L, M, T) are involved, select three fundamental variables, while if the problem involves only two fundamental dimensions, two fundamental variables are needed.
- 2) Important variables should be selected as fundamental variables. In other words, do not choose secondary variables as fundamental variables; otherwise secondary variables appear in most terms, which often complicates the problem or even requires re-solving.

⇒ 例题1

⇒ 例题2

⇒ 例题3

⇒ 例题4



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Example 1: Liquid flows in a horizontal equal-diameter tube. Let the pressure difference at two points Δp be related to these variables: pipe diameter d , ρ , v , l , μ , pipe wall roughness Δ . Find the formulation for Δp .

1) General form of the formulation: $f(d, \rho, \mu, v, l, \Delta, \Delta p) = 0$

2) Fundamental variables: d, ρ, v

3) π numbers: $N(\pi) = n - m = 7 - 3 = 4$. The remaining variables are μ, l, Δ and Δp .

$$\pi_1 = d^{a_1} v^{b_1} \rho^{c_1} \cdot \mu; \quad \pi_2 = d^{a_2} v^{b_2} \rho^{c_2} \cdot l$$

$$\pi_3 = d^{a_3} v^{b_3} \rho^{c_3} \cdot \Delta; \quad \pi_4 = d^{a_4} v^{b_4} \rho^{c_4} \cdot \Delta p$$

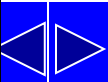
4) Solving for indexes of π numbers

$$\dim \pi_1 = \dim(d^{a_1} v^{b_1} \rho^{c_1} \cdot \mu)$$

$$M^0 L^0 T^0 = [L]^{a_1} [LT^{-1}]^{b_1} [MT^{-3}]^{c_1} [ML^{-1}T^{-1}]$$

$$\left. \begin{aligned} M : 0 &= c_1 + 1 \\ L : 0 &= a_1 + b_1 - 3c_1 - 1 \\ T : 0 &= -b_1 - 1 \end{aligned} \right\} \therefore \begin{cases} a_1 = -1 \\ b_1 = -1 \\ c_1 = -1 \end{cases} \therefore \pi_1 = \frac{1}{\text{Re}}$$

Similarly, $\pi_2 = \frac{l}{d}; \quad \pi_3 = \frac{\Delta}{d}; \quad \pi_4 = \frac{\Delta p}{\rho v^2}$



Example 1: Liquid flows in a horizontal equal-diameter tube. Let the pressure difference at two points Δp be related to these variables: pipe diameter d , ρ , v , l , μ , pipe wall roughness Δ . Find the formulation for Δp .

5) Determine the formulation for Δp

$$f_1(\pi_1, \pi_2, \pi_3, \pi_4) = f_1\left(\frac{l}{d}, \frac{1}{Re}, \frac{\Delta}{d}, \frac{\Delta p}{\rho v^2}\right) = 0$$

$$\frac{\Delta p}{\rho v^2} = f_2\left(\frac{l}{d}, Re, \frac{\Delta}{d}\right)$$

Because of $\Delta p \propto l$, we have

$$\Delta p = f_3\left(Re, \frac{\Delta}{d}\right) \frac{l}{d} \rho v^2$$

$$\Delta h = \frac{\Delta p}{\rho g} = f_4\left(Re, \frac{\Delta}{d}\right) \frac{l}{d} \frac{v^2}{2g}$$

$$h_f = \lambda \frac{l}{d} \frac{v^2}{2g} \text{ and } \lambda = f_4\left(Re, \frac{\Delta}{d}\right)$$

Example 2: Establish the formulation for flow resistance of viscous fluid passing through a vertical long straight cylinder of unit length. 20

1) **Physical variables related to the problem:** Resistance per unit length $F_D = F/L$ (F is the total resistance force on the column; L is the column length). Factors affecting resistance include column diameter D , fluid density ρ , viscosity μ , and flow rate U . The function reads:

$$f(F_D, D, U, \rho, \mu) = 0$$

2) **Fundamental variables:** ρ, U, D . $n=5, m=3$.

Remaining variables are F_D and μ .

3) π numbers: $n-m=2$ $\pi_1 = \rho^{a_1} U^{b_1} D^{c_1} \cdot F_D$; $\pi_2 = \rho^{a_2} U^{b_2} D^{c_2} \cdot \mu$

4) **Indexes of π numbers**

$$\dim \pi_1 = \dim(\rho^{a_1} U^{b_1} D^{c_1} \cdot F_D)$$

$$M^0 L^0 T^0 = [ML^{-3}]^{a_1} [LT^{-1}]^{b_1} [L]^{c_1} [MLT^{-2}L^{-1}]$$

Similarly,

$$\left. \begin{aligned} M: 0 &= a_1 + 1 \\ L: 0 &= -3a_1 + b_1 + c_1 \\ T: 0 &= -b_1 - 2 \end{aligned} \right\} \therefore \begin{cases} a_1 = -1 \\ b_1 = -2 \\ c_1 = -1 \end{cases} \therefore \pi_1 = \frac{F_D}{\rho D U^2} \quad \pi_2 = \frac{\mu}{\rho D U} = Re^{-1}$$

Example 2: Establish the formulation for flow resistance of viscous fluid passing through a vertical long straight cylinder of unit length. 21

5) Establish the formulation

$$f\left(\frac{F_D}{\rho DU^2}, Re\right) = 0$$

Or solve π_1

$$F_D = f_1(Re) \cdot \rho DU^2 = C_D \rho DU^2$$

Recall

Substitute all π terms into the π —theorem function

$$\pi_1 = \frac{F_D}{\rho DU^2}$$

$$\pi_2 = \frac{\mu}{\rho DU} = Re^{-1}$$

$$f(\pi_1, \pi_2, \dots, \pi_{n-m}) = 0$$

meter, such as:

$$\pi_4 = f(\pi_1, \pi_2, \dots, \pi_{n-m})$$

action for the dependent variable.



Example 3: Venturi flowmeters can measure the flow rate of pressurized pipelines.

Pressure drop Δp is related to flow rate q_V , fluid density ρ , liquid viscosity coefficient μ , tube wall roughness height Δ , flowmeter length L and pipe diameters D_1 and D_2 .

Using π theorem, solve for the flow rate formula expressed by pressure drop Δp .

Solution:

1) General form of the formulation:

$$f(\rho, q_V, D_1, \mu, D_2, \Delta p) = 0$$

2) Fundamental variables: ρ, q_V, D_1 ; $n=6, m=3$

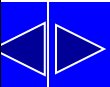
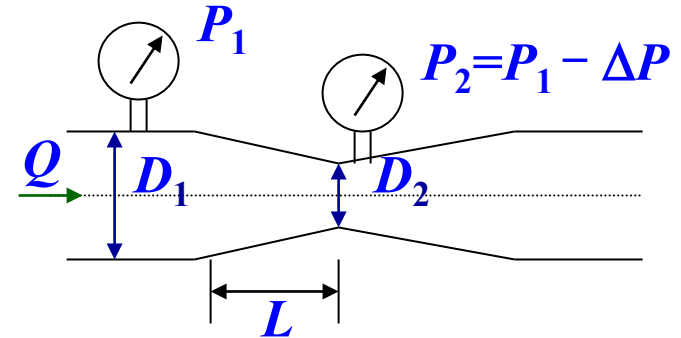
3) Number of π numbers: $N(\pi) = n - m = 6 - 3 = 3$

4) Solving for indexes of π numbers

$$\pi_1 = \rho^{\alpha_1} q_V^{\beta_1} D_1^{\gamma_1} \cdot \Delta P$$

$$\pi_2 = \rho^{\alpha_2} q_V^{\beta_2} D_1^{\gamma_2} \cdot \mu$$

$$\pi_3 = \rho^{\alpha_3} q_V^{\beta_3} D_1^{\gamma_3} \cdot D_2$$



Express the π numbers by fundamental dimensions:

$$[\pi_1] = [ML^{-3}]^{\alpha_1} [L^3T^{-1}]^{\beta_1} [L]^{\gamma_1} [ML^{-1}T^{-2}]$$

$$\therefore \alpha_1 = -1, \quad \beta_1 = -2, \quad \gamma_1 = 4$$

$$\pi_1 = \left(\frac{q}{D_1^2} \sqrt{\rho / \Delta P} \right)^{-2}$$

Similarly :

$$\pi_2 = \left(\frac{q\rho}{D_1\mu} \right)^{-1} \quad \pi_3 = \frac{D_2}{D_1}$$

5) Write out the expression for flow rate

$$\frac{q}{D_1^2} \sqrt{\rho / \Delta P} = f_1 \left(\frac{q\rho}{D_1\mu}, \frac{D_2}{D_1} \right)$$

$$\text{or: } q = D_1^2 \sqrt{\frac{\Delta p}{\rho}} f(Re, \frac{D_2}{D_1})$$

Example 4: Expression of wall shear stress τ_0 in pipe flow determined by Buckingham Theorem. Given that τ_0 is related to liquid density ρ , dynamic viscosity of liquid μ , tube diameter D , the pipe wall roughness Δ , and the average flow velocity v in the pipe.

1) General form of the formulation: $f(D, v, \rho, \tau_0, \mu, \Delta) = 0$

2) Fundamental variables: D, v, ρ ; $n=6, m=3$

3) Number of π numbers: $N(\pi) = n - m = 6 - 3 = 3$

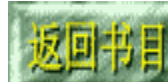
4) Solving for indexes of π numbers

$$\pi_1 = \frac{\tau_0}{D^{a_1} v^{b_1} \rho^{c_1}}, \pi_2 = \frac{\mu}{D^{a_2} v^{b_2} \rho^{c_2}}, \pi_3 = \frac{\Delta}{D^{a_3} v^{b_3} \rho^{c_3}}$$

Express the π numbers by fundamental dimensions. For π_1 :

$$[ML^{-1}T^{-2}] = [L]^{a_1} [LT^{-1}]^{b_1} [ML^{-3}]^{c_1}$$

$$\left. \begin{array}{l} L : -1 = a_1 + b_1 - 3c_1 \\ T : -2 = -b_1 \\ M : 1 = c_1 \end{array} \right\} \therefore \begin{cases} a_1 = 0 \\ b_1 = 2 \\ c_1 = 1 \end{cases} \therefore \pi_1 = \frac{\tau_0}{\rho v^2}$$



Similarly, $\pi_2 = \frac{\mu}{Dv\rho} = \frac{1}{Re}$, $\pi_3 = \frac{\Delta}{D}$

5) Write out the expression

$$f\left(\frac{\tau_0}{\rho v^2}, \frac{1}{Re}, \frac{\Delta}{D}\right) = 0$$

Substitute all π terms into the π —theorem function

$$f(\pi_1, \pi_2, \dots, \pi_{n-m}) = 0$$

meter, such as:

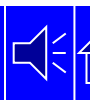
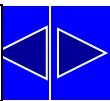
$$\pi_4 = f(\pi_1, \pi_2, \dots, \pi_{n-m})$$

action for the dependent variable.

Re-arranging gives

$$\frac{\tau_0}{\rho v^2} = f\left(\frac{1}{Re}, \frac{\Delta}{D}\right)$$

Letting $f\left(\frac{1}{Re}, \frac{\Delta}{D}\right) = \frac{\lambda}{8}$, we get $\tau_0 = \frac{\lambda}{8} \rho v^2$



(1) The role of dimensional analysis?

- Deriving the dimension of a physical variable;
- Simplifying physical equations;
- Verifying the correctness and completeness of physical equations and empirical formulas;
- Providing theoretical guidance for planning experiments and processing experimental data.

(2) Whether empirical formulas satisfy the principle of dimensional homogeneity?

Generally not satisfied, because a empirical formula is usually based on the statistics of a series of experimental data, without considering the dimensional homogeneity.

(3) What are the applicable situations for the Rayleigh Method and the Buckingham π -Theorem?

Rayleigh method is suitable for relatively simple problems with less than 4-5 variables, while the π -theorem is a more general method.

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Chapter 5 Dimensional analysis and similarity principle

Section 3 Basis of flow similarity theory

- ➡ 1. Conditions of similarity
- 2. Derivation of similarity criterion numbers
- 3. Physical meaning of similarity criterion numbers
- 4. Dynamic similarity criterion
- 5. Similarity criterion number for compressible flow

1. Conditions of similarity

Flow similarity
between model and
prototype requires
the satisfaction of
these conditions:

- Geometric similarity
- Kinematic similarity
- Dynamic similarity
- Similarity of initial conditions and boundary conditions

(1) Geometric similarity

- The flow fields of prototype and model are similar in geometry.
- That is, the ratios of the linear variables of prototype and model and their flow motions are equal, as well as the corresponding angles are equal.

Scale of length: $\lambda_L = \frac{l_p}{l_m}$

Scale of area: $\lambda_A = \frac{A_p}{A_m} = \frac{l_p^2}{l_m^2} = \lambda_l^2$

Scale of volume: $\lambda_V = \frac{V_p}{V_m} = \frac{l_p^3}{l_m^3} = \lambda_l^3$

(2) Kinematic similarity

- The velocity fields of fluid motions are similar
- That is, the velocities u and accelerations a at corresponding locations (including those on boundaries) of model and prototype flows are in the same direction, and their magnitudes follow a fixed ratio.

$$\text{Scale of velocity } \lambda_v = \frac{v_p}{v_m} = \frac{(l/t)_p}{(l/t)_m} = \lambda_l \lambda_t^{-1}$$

$$\text{Scale of acceleration: } \lambda_a = \frac{a_p}{a_m} = \frac{(v/t)_p}{(v/t)_m} = \lambda_v \lambda_t^{-1} = \lambda_l \lambda_t^{-2}$$

(3) Dynamic similarity

- The forces of the same type applied on the corresponding locations of model and prototype flows are in the same direction, and their magnitudes follow a fixed ratio.

$$\lambda_F = \frac{F_p}{F_m}$$

If gravity is the only mass force, we have:

$$\lambda_f = \frac{f_p}{f_m} = \frac{g_p}{g_m} = \lambda_g$$

(4) Similarity of initial conditions and boundary conditions

Initial conditions: Suitable for unsteady flow

Boundary conditions: If flow similarity is true, the similarity of boundary conditions is satisfied automatically

Implications of flow similarity:

- Geometric similarity is the premise and basis of kinematic similarity and dynamic similarity;
- Dynamic similarity is the dominant factor for the kinematic similarity of two fluid flows;
- Kinematic similarity is the manifestation of geometric similarity and dynamic similarity;
- If flows are similar, they are flows satisfying geometric similarity, kinematic similarity and dynamic similarity



Chapter 5 Dimensional analysis and similarity principle

Section 3 Basis of flow similarity theory

1. Conditions of similarity
2. Derivation of similarity criterion numbers
3. Physical meaning of similarity criterion numbers
4. Dynamic similarity criterion
5. Similarity criterion number for compressible flow

2. Derivation of similar criterion numbers

N-S equation with
gravity being the
only mass force

$$\underbrace{\vec{f}}_{\text{Gravity}} - \underbrace{\frac{1}{\rho} \nabla p}_{\text{Pressure}} + \underbrace{\nu \nabla^2 \vec{u}}_{\text{Viscous force}} = \underbrace{\frac{\partial \vec{u}}{\partial t}}_{\text{Time-varying acceleration inertia force}} + \underbrace{(\vec{u} \cdot \nabla) \vec{u}}_{\text{Convective acceleration inertia force}}$$

Convert the equation into the dimensionless form.

For the **model**

$$\left(\underbrace{\frac{\vec{f}_m}{(\vec{u} \cdot \nabla) \vec{u}}}_{\text{Gravity Inertia force}} - \underbrace{\frac{1}{\rho} \frac{\nabla p}{(\vec{u} \cdot \nabla) \vec{u}}}_{\text{Pressure Inertia force}} + \underbrace{\frac{\nu \nabla^2 \vec{u}}{(\vec{u} \cdot \nabla) \vec{u}}}_{\text{Viscous force Inertia force}} = \underbrace{\frac{\partial \vec{u} / \partial t}{(\vec{u} \cdot \nabla) \vec{u}}}_{\text{Time-varying acceleration inertia force}} + 1 \right)_m \quad (1)$$

For the **prototype**, where $\nabla = \frac{\partial}{\partial x}$, $\lambda_\nabla = \frac{1}{\lambda_l}$

$$\frac{\lambda_g \lambda_l}{\lambda_u^2} \left(\frac{\vec{f}_m}{(\vec{u} \cdot \nabla) \vec{u}} \right) - \frac{\lambda_p}{\lambda_\rho \lambda_u^2} \left(\frac{1}{\rho} \frac{\nabla p}{(\vec{u} \cdot \nabla) \vec{u}} \right) + \frac{\lambda_\nu}{\lambda_l \lambda_u} \left(\frac{\nu \nabla^2 \vec{u}}{(\vec{u} \cdot \nabla) \vec{u}} \right) = \frac{\lambda_l}{\lambda_t \lambda_u} \left(\frac{\partial \vec{u} / \partial t}{(\vec{u} \cdot \nabla) \vec{u}} \right) + 1 \quad (2)$$

If two flows are similar, Eqs. (1) and (2) should equal. The condition of the equality is that the coefficients of the dimensions of each term in Eq. (2) equal to 1. That is,

$$\underbrace{\frac{\lambda_g \lambda_l}{\lambda_u^2} \left(\frac{\vec{f}_m}{(\vec{u} \cdot \nabla) \vec{u}} \right)}_{\substack{\text{Gravity} \\ \text{Inertia force}}} - \underbrace{\frac{\lambda_p}{\lambda_\rho \lambda_u^2} \left(\frac{1}{\rho} \frac{\nabla p}{(\vec{u} \cdot \nabla) \vec{u}} \right)}_{\substack{\text{Pressure} \\ \text{Inertia force}}} + \underbrace{\frac{\lambda_v}{\lambda_l \lambda_u} \left(\frac{\nu \nabla^2 \vec{u}}{(\vec{u} \cdot \nabla) \vec{u}} \right)}_{\substack{\text{Viscous force} \\ \text{Inertia force}}} = \underbrace{\frac{\lambda_l}{\lambda_t \lambda_u} \left(\frac{\partial \vec{u} / \partial t}{(\vec{u} \cdot \nabla) \vec{u}} \right) + 1}_{\substack{\text{Time-varying acceleration inertia force} \\ \text{Migration acceleration inertia force}}}$$

$$\frac{\lambda_g \lambda_l}{\lambda_u^2} = \frac{\lambda_p}{\lambda_\rho \lambda_u^2} = \frac{\lambda_v}{\lambda_l \lambda_u} = \frac{\lambda_l}{\lambda_t \lambda_u} = 1$$

Therefore $\frac{v_p^2}{g_p l_p} = \frac{v_m^2}{g_m l_m} ; \frac{p_p}{\rho_p v_p^2} = \frac{p_m}{\rho_m v_m^2} ; \frac{v_p l_p}{\nu_p} = \frac{v_m l_m}{\nu_m} ; \frac{l_p}{v_p t_p} = \frac{l_m}{v_m t_m}$

or $Fr_p = Fr_m ; Eu_p = Eu_m ; Re_p = Re_m ; St_p = St_m$

Froude Number Euler Number Reynolds Number Strouhal Number

$$\underbrace{\frac{\text{Inertia force}}{\text{Gravity}}}_{\text{Gravity}} \quad \underbrace{\frac{\text{Pressure}}{\text{Inertia force}}}_{\text{Inertia force}} \quad \underbrace{\frac{\text{Inertia force}}{\text{Viscous force}}}_{\text{Viscous force}} \quad \underbrace{\frac{\text{Time-varying acceleration inertia force}}{\text{Migration acceleration inertia force}}}_{\text{Migration acceleration inertia force}}$$

Generally defined as: $Fr = \frac{v}{\sqrt{gl}} \quad Eu = \frac{\Delta p}{\rho v^2}$

Chapter 5 Dimensional analysis and similarity principle

Section 3 Basis of flow similarity theory

1. Conditions of similarity
2. Derivation of similarity criterion numbers
-  3. Physical meaning of similarity criterion numbers
4. Dynamic similarity criterion
5. Similarity criterion number for compressible flow

3. Physical meaning of similarity criterion numbers

In physical model test, it is crucial to figure out the primary forces affecting fluid motion and adopt the similarity criteria to design experiments. Therefore, it is necessary to understand the physical meanings of similarity numbers.

$$\underbrace{\vec{f}}_{\text{Gravity}} - \underbrace{\frac{1}{\rho} \nabla p}_{\text{Pressure}} + \underbrace{\nu \nabla^2 \vec{u}}_{\substack{\text{Viscous} \\ \text{force}}} = \underbrace{\frac{\partial \vec{u}}{\partial t}}_{\substack{\text{Time-varying} \\ \text{acceleration} \\ \text{inertia force}}} + \underbrace{(\vec{u} \square \nabla) \vec{u}}_{\substack{\text{Convective acceleration} \\ \text{inertia force}}}$$

- Among the various forces applied on fluids, their influences on fluid motions are different (some stronger and some weaker). In model tests, it is necessary to determine the force that plays a leading role in flow motion according to the real situation.
- In addition to the inertial force, all other forces attempt to change the state of motion. Therefore, it is a good option to establish the proportional relationships of these forces with the inertial force.



3. Physical meaning of similarity criterion numbers (Cont'd)

	Physical meaning	Implications
Fr (弗劳德数)	(Ratio of) inertial force of fluid to gravity	Froude numbers of two flows being equal implies the gravity being similar
Eu (欧拉数)	Pressure to inertial force	Euler numbers being equal implies pressures being similar
Re (雷诺数)	Inertial force to viscous force	Reynolds numbers being equal implies viscous forces being similar
St (斯特劳哈尔数)	Time-varying acceleration inertia force to convective acceleration inertia	Strouhal numbers being equal implies time-varying acceleration inertia force being similar

For two dynamically similar flows of viscous incompressible fluids, each of the four similarity criterion numbers, i.e. Fr, Eu, Re, St, needs to be equal. This implies that the ratios of inertia forces, gravity forces, pressures, viscous forces acting on arbitrary corresponding locations in the two flow fields need to be equal.



Chapter 5 Dimensional analysis and similarity principle

Section 3 Basis of flow similarity theory

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4. Dynamic similarity criterion

The dynamic similarity condition (i.e. similarity criterion numbers being equal) is called the dynamic similarity criterion. The complete dynamic similarity requires each similarity criterion number being equal, but in practice, this is often impossible to achieve.

For example, water is used as a model fluid to investigate a prototype water flow. The equalities of Froude number and Reynolds number give

$$\text{With } (Re)_p = (Re)_m \quad \text{or} \quad \lambda_v \cdot \lambda_l / \lambda_v = 1, \lambda_v = 1 \quad \text{We have } \lambda_v = \lambda_l^{-1}$$

$$\text{With } (Fr)_p = (Fr)_m, \text{ and } \lambda_g = 1 \quad \text{We have } \lambda_v = \lambda_l^{1/2}$$

$$Fr = \frac{v}{\sqrt{gl}}$$

$$\text{Then we can get } \lambda_l^{1/2} = \lambda_l^{-1}$$

The above equation hold only when $\lambda_l = 1$. Obviously, this has lost the intension of model tests with similarity.

4. Dynamic similarity criterion (Cont'd)

- Usually, we select one force dominating the flow such that the corresponding similarity criterion number is equal;
- If the flows are dominated by gravity, then let the ***Fr*** number be equal, which is called the Fr number similarity criterion, or the gravity similarity criterion;
- If the flows are dominated by viscous force, let the ***Re*** number be equal, which is called the Re similarity criterion, or the viscous force (resistance) similarity criterion;
- The others include the Euler similarity criterion (pressure similarity criterion) and so on. In engineering fluid mechanics, Fr similarity criterion and Re similarity criterion are the most widely used.



Chapter 5 Dimensional analysis and similarity principle

Section 3 Basis of flow similarity theory

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5. Extension - Similarity criterion number for compressible fluids

For compressible fluids, when the elastic force matters, two more similarity criteria numbers can be obtained from the proportional relation of forces, i.e.

Cauchy number and Mach number.

(1) Cauchy number $\left(\frac{f}{f_E}\right)_p = \left(\frac{f}{f_E}\right)_m$

- The elastic force of fluid per unit mass is $f_E = \frac{Kl^2}{\rho l^3} = \frac{K}{\rho l}$

where K is the bulk modulus of fluid, and its unit is N/m².

- Inertia force of unit mass fluid $f = a = \frac{v^2}{l}$
- We can get $\left(\frac{\rho v^2}{K}\right)_p = \left(\frac{\rho v^2}{K}\right)_m$

Notes: The Cauchy number indicates the **ratio of inertial force to elastic force**.

The Cauchy numbers of two flows being equal implies the elastic force similarity.

Cauchy similarity criterion is used for studying the water hammer phenomenon.



5. Extension - Similarity criterion number for compressible fluids

(2) Mach number

- Speed of sound in fluid $c = \sqrt{\frac{K}{\rho}}$
- Substitute c into the formula for Cauchy number : $(\frac{\rho v^2}{K})_p = (\frac{\rho v^2}{K})_m$

We get: $\frac{v_p}{c_p} = \frac{v_m}{c_m}$ or $(Ma)_p = (Ma)_m$

- The dimensionless number $Ma = v/c$ is called the Mach number.
- When the velocity of compressible airflow is close to or exceeds the speed of sound, the elastic force becomes the dominant factor affecting the flow.
- Then the equality of the Mach number is needed for flow similarity.



Chapter 5 Dimensional analysis and similarity principle

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Chapter 5 Dimensional analysis and similarity principle

Section 4 Model experiment

-  1. Selection of model similarity criteria
- 2. Model design
- 3. Data processing of model experiment

1. Selection of model similarity criteria

Reynolds number criterion——Pressurized pipe flow, flow around a submerged body with a low Re number, etc.

Gravity similarity criterion——Open channel flow, weir flow, flow around bridge piers and the local flow in a pressurized pipe with vortexes, **wave impact on structures**, etc.

Notes

- Not all flows cannot achieve both the viscous force similarity and the gravity similarity at the same time.
- When Re number exceeds a certain value, the flow resistance coefficient becomes a constant not varying with Re.
- When both the prototype and model are in this flow regime, the model can be designed based on the gravity similarity criterion. In this way, the viscous force similarity is realized automatically without the equality of Re (resistance coefficients being equal).



Chapter 5 Dimensional analysis and similarity principle

Section 4 Model experiment

1. Selection of model similarity criteria



2. Model design

3. Data processing of model experiment

2 Model design

Model design generally follows these steps:

- (1) Determine the length scale
- (2) Determine the geometric size of the model
- (3) Select the similarity criterion
- (4) Calculate similarity scales of all physical variables of the flow
- (5) Calculate flow rate, velocity and quantities of the model based on scale, and determine boundary conditions



Chapter 5 Dimensional analysis and similarity principle

Section 4 Model experiment

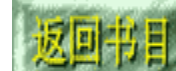
1. Selection of model similarity criteria
2. Model design
3. Data processing of model experiment



3. Data process of model experiment

Data of models satisfying similarity can be generally divided into two categories:

- One is the dimensional quantity, e.g. flow resistance, pressure, velocity distribution, etc.
 - The other is the quantity of dimension one, e.g. flow resistance coefficient of flow past a cylinder C_D
- For quantities of dimension one, because the model and the prototype flows are similar, the model values are equal to those of the prototype, and there is no need to convert values.
- For the dimensional quantities, need to converted values of model tests to prototype values, based on the scales of the selected similarity criterion.



Example 1: There is an oil pipeline with a diameter of 15 cm, a length of 5 m, and a required flow rate of $0.18 \text{ m}^3/\text{s}$. Model tests are carried out using water. The model pipe diameter is the same as the prototype and the water temperature is 10°C ; the kinematic viscosity coefficient of oil in prototype is $\nu_p = 0.13 \text{ cm}^2/\text{s}$.

(1) Calculate the flow rate of water in model test so as to achieve similarity.

Solution: Because the flow in a circular tube is mainly influenced by viscous force, the Reynolds similarity criterion needs to be satisfied. That is, the Reynolds numbers of prototype and model are equal,

$$\frac{v_p d_p}{\nu_p} = \frac{v_m d_m}{\nu_m}$$

Since $d_p = d_m$, the above formula can be written as

$$\frac{v_p}{\nu_p} = \frac{v_m}{\nu_m}, \quad v_m = \frac{\nu_m}{\nu_p} v_p \quad \text{or} \quad q_{v_m} = \frac{\nu_m}{\nu_p} q_{v_p}$$

Substituting the known values $\nu_p = 0.13 \text{ cm}^2/\text{s}$, $\nu_m = 0.0131 \text{ cm}^2/\text{s}$ gives

$$q_{v_m} = \frac{0.0131}{0.13} \times 0.18 = 0.0181 \text{ m}^3/\text{s}$$

That is, when the flow rate in the model is $q_{v_m} = 0.0181 \text{ m}^3/\text{s}$, the flows in prototype and model are similar.

Example 1: There is an oil pipeline with a diameter of 15 cm, a length of 5 m, and a required flow rate of $0.18 \text{ m}^3/\text{s}$. Model tests are carried out using water. The model pipe diameter is the same as the prototype and the water temperature is 10°C ; the kinematic viscosity coefficient of oil in prototype is $\nu_p = 0.13 \text{ cm}^2/\text{s}$.

(2) If the pressure difference between the two ends of a 5 m-long model pipeline is 3 cm, calculate the pressure difference between the two ends of a 5m-long oil pipeline (expressed by the height of the oil column).

Solution: With the Reynolds similarity criterion holding, the Euler number criterion is automatically satisfies, i.e.

$$Eu = \frac{\Delta p}{\rho v^2}$$

$$\left(\frac{\Delta p}{\rho v^2}\right)_p = \left(\frac{\Delta p}{\rho v^2}\right)_m$$

Reorganization gives: $\frac{\Delta p_p}{\rho g_p} = \frac{\Delta p_m}{\rho g_m} \cdot \frac{v_p^2}{v_m^2} \cdot \frac{g_m}{g_p}$

Since $A_p = A_m$ ($d_p = d_m$) and $g_p = g_m$, we have $\frac{\Delta p_p}{\rho g_p} = \frac{\Delta p_m}{\rho g_m} \cdot \frac{q_{Vp}^2}{q_{Vm}^2}$

Therefore, the pressure difference in a 5m-long oil pipeline is:

$$h_p = \frac{\Delta p_p}{\rho g_p} = h_m \frac{q_{Vp}^2}{q_{Vm}^2} = 0.03 \times \frac{0.18^2}{0.0181^2} = 2.95 \text{ m}$$



Example 2: A ship model with a length scale $\lambda_L = 50$ is sailing at a speed of 1m/s in a towing tank. The wave resistance is measured to be 0.02 N.

(1) Solve for the wave resistance in prototype; (2) Solve for the ship sailing velocity in prototype; (3) Calculate the power required in prototype.

Solution: Gravity plays a dominant role. So the Froude numbers of the prototype and the model should equal. That is:

$$\frac{v_p^2}{g_p l_p} = \frac{v_m^2}{g_m l_m}$$

$$Fr = \frac{v}{\sqrt{gl}}$$

Since $g_p = g_m$, the above formula can be written as:

$$\frac{v_p^2}{l_p} = \frac{v_m^2}{l_m} \quad \text{or} \quad \frac{l_p}{l_m} = \frac{v_p^2}{v_m^2} = \lambda_l = 50$$

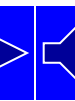
$$\begin{aligned} \text{Since } \frac{F_p}{F_m} &= \frac{\rho_p l_p^2 v_p^2}{\rho_m l_m^2 v_m^2} = \lambda_\rho \lambda_l^2 \lambda_v^2 \\ &= \lambda_\rho \cdot \lambda_l^2 \cdot \lambda_l = \lambda_l^3 \end{aligned}$$

$$v_p = \lambda_l^{\frac{1}{2}} v_m = 50^{\frac{1}{2}} \times 1 = 7.1 \text{ m/s}$$

$$N_p = F_p v_p = 2500 \times 7.1$$

$$\text{So } F_p = \lambda_l^3 F_m = 50^3 \times 0.02 = 2500 \text{ N}$$

$$= 17800 \text{ N} \cdot \text{m/s} = 17.8 \text{ kW}$$



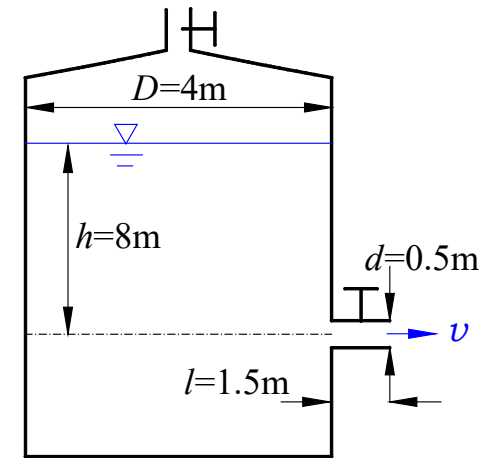
Example 3: In an oil tank, the oil temperature is $T = 20^\circ\text{C}$, the kinematic viscosity coefficient of oil is $\nu_p = 0.74 \text{ cm}^2/\text{s}$. Taking the length scale λ_L around 4, do the following studies: (1) Which similarity criteria needs to be selected? (2) Select the fluid in model tests? (3) Calculate the scales of variables.

Solution: (1) Oil flows out from the tube and is under the action of gravity. Due to the large viscosity of oil, it is also affected by the viscous force. Therefore, both gravity and viscous forces are important forces. So the similarity criterions here should satisfy both Reynolds number and Froude number.

$$(2) \quad \because \lambda_{Re} = \lambda_{Fr}; \quad \frac{\lambda_v \cdot \lambda_l}{\lambda_v} = \frac{\lambda_v}{\lambda_g^{\frac{1}{2}} \lambda_l^{\frac{1}{2}}}$$

$$\therefore \lambda_v = \lambda_l^{\frac{3}{2}} = 4^{\frac{3}{2}} = 8$$

$$\nu_m = \frac{\nu_p}{\lambda_v} = \frac{0.74}{8} = 0.0925 \text{ cm}^2/\text{s}$$



- Fluids with $\nu = 0.0925 \text{ cm}^2/\text{s}$ are difficult to find. Need to choose fluids with close viscosities.
- Choose 59% glycerol solution at 20°C . $\nu = 0.0892 \text{ cm}^2/\text{s}$. Close to the above calculated value, but the temperature should be maintained in 20°C during tests.
- In this temperature, ν of the model liquid is $0.0892 \text{ cm}^2/\text{s}$ but not $0.0925 \text{ cm}^2/\text{s}$.



(3) After the model fluid is selected, the length scale λ_l should be corrected because the selected ν_m is no longer $0.0925 \text{ cm}^2/\text{s}$

$$\lambda_l = \lambda_v^{\frac{2}{3}} = \left(\frac{\nu_p}{\nu_m}\right)^{\frac{2}{3}} = \left(\frac{0.74}{0.0892}\right)^{\frac{2}{3}} = 4.1$$

That is, the length scale λ_l is 4.1 but not 4. Therefore, the diameter of the model tank is:

$$D_m = \frac{D_p}{\lambda_l} = \frac{4}{4.1} = 0.976 \text{ m} \quad h_m = \frac{8}{4.1} = 1.95 \text{ m}$$

$$d_m = \frac{0.5}{4.1} = 0.122 \text{ m} \quad d_m = \frac{1.5}{4.1} = 0.366 \text{ m}$$

Velocity scale λ_v is obtained based on the Froude number criterion (the same result can be obtained from the Reynolds number criterion):

$$Fr = \frac{v}{\sqrt{gl}}$$

$$\lambda_v = \lambda_l^{\frac{1}{2}} = 4.1^{\frac{1}{2}} = 2.025$$

This means that the flow rate in the model pipe is around half of that in the prototype.

Time scale λ_t can be obtained from the Reynolds number criterion

$$\lambda_t = \lambda_l \lambda_v^{-1} = \lambda_l \lambda_l^{-\frac{1}{2}} = \lambda_l^{\frac{1}{2}} = 2.025$$

Acceleration scale $\lambda_a = \frac{\lambda_v}{\lambda_t} = 1$

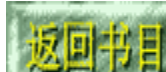


Q1. Why does each similarity criterion relate with the inertial force?

- The forces acting on fluids are those attempting to change the state of motion except the inertial force that maintains fluid's original state of motion
- If the forces acting on a fluid form a force polygon, the inertial force is the resultant force of the force polygon, i.e. the Newton's Second Law $\sum \vec{F} = m\vec{a}$
- The change of flow motion is the result of the interaction between inertial force and other forces

Q2. Exemplify the water flows dominated by gravity and viscous force

- Viscous force: pipe flows and pressurized flows in tunnels with $Re < 2000$, flow past submerged objects, etc.
- Gravity : Weir overflow, orifice outflow, open channel flow, and pressurized flows in pipes/tunnels in the resistance squared zone, wave impact on structures.



Q3. Can the prototype and the model satisfy both the gravity similarity criterion and the viscous force similarity criterion? Why?

- If the same fluid is used, the answer is no, because of the gravity similarity.

$$\frac{\lambda_v}{\lambda_l^{1/2} \cdot \lambda_g^{1/2}} = 1 \therefore \lambda_v = \lambda_l^{1/2}$$

Similar viscous force $\frac{\lambda_v \lambda_l}{\lambda_v} = 1 \therefore \lambda_v = \lambda_l^{-1}$

- If using different fluids, the answer is yes theoretically.

Gravity similarity $\frac{\lambda_v}{\lambda_l^{1/2} \cdot \lambda_g^{1/2}} = 1 \therefore \lambda_v = \lambda_l^{1/2}$

Viscous force similarity $\frac{\lambda_v \lambda_l}{\lambda_v} = 1 \therefore \lambda_v = \lambda_l^{-1} \cdot \lambda_v$

$\lambda_v = \lambda_l^{\frac{3}{2}}$

However, it is difficult to obtain $\lambda_v = \lambda_l^{\frac{3}{2}}$ in practice.

Chapter 5 Dimensional analysis and similarity principle

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Chapter Summary

(1) If two liquid flows are similar, the following aspects need to be satisfied:

- Geometric similarity
- Kinematic similarity
- Dynamic similarity
- Both initial conditions and boundary conditions being similar

(2) Similarity criteria: Re similarity criterion, Fr similarity criterion, Eu similarity criterion

a. When the viscous force plays a major role $Re = \frac{\text{Viscosity force}}{\text{inertia force}} = \frac{\nu l}{\nu}$

b. When gravity plays a major role, $Fr = \frac{\text{Inertia force}}{\text{Gravity}} = \frac{v}{\sqrt{gL}}$

c. When pressure plays a major role, $Eu = \frac{\text{Pressure}}{\text{Inertia force}} = \frac{\Delta p}{\rho v^2}$

When the same fluid is used in prototype and model, gravity similarity and viscous force similarity cannot be satisfied at the same time. So, given the viscous force or gravity at the corresponding point is similar, the pressure will be similar.

(3) Fundamental dimension: independent dimensions that cannot be derived from other dimensions. Generally, [L-M-T] is taken.

Derived dimension: Dimensions derived from fundamental dimensions, e.g. $[X]=[L^\alpha, T^\beta, M^\gamma]$.

(4) Principle of dimensional homogeneity

(5) Two methods of dimensional analysis

- Rayleigh method
- Buckingham π theorem

(6) Flow similarity theory

- Similar numbers and their physical meanings
- Dynamic similarity criterion

(7) Model experiment

- Selection of similarity criteria
- Model design
- Processing of experimental data